

VERIFIED RESULTS · REFRIGERATED DISTRIBUTION CENTER

Patented capital equipment cut refrigeration energy **26.5%** at a refrigerated distribution center.

Validated by ON/OFF metering. Sub-12-month payback. Zero cold-chain disruption.

26.5%

REFRIGERATION REDUCTION

112,432

KWH ANNUALIZED

66,335^{kg}CO₂ ABATED/YR**~11^{mo}**

PROJECTED PAYBACK

SOLUTION

Patented enPact Refrigeration & Cooling optimization. Installed across all 5 cold rooms and 3 freezer systems. Reduces compressor runtime (80–90% of refrigeration energy) via superheat-based control and stabilizes setpoints without modifying site operations.

- Reduce compressor runtime and cycling
- Regulate refrigerant flow via superheat control
- Increase latent cooling efficiency
- Maintain setpoint temperatures without modification
- Verified via smart kWh meters + M&V (Univers platform)

CHALLENGE

A national food distribution center with a large refrigerated warehouse faced rising energy consumption. Refrigeration represented the majority of the site's electrical usage. Key constraints:

- Freezers operating at 68.5 kW each; cold rooms 15–19 kW
- 25% seasonal uplift during warmer months
- Transparent, M&V-validated savings required before wider rollout
- Sub-12-month payback required for investment approval
- No option for cold-chain disruption

Results.***Whole-warehouse refrigeration savings.***

Measured over 34 days (Feb–Mar) using ON/OFF comparison windows. Full-year figure adjusts for the 25% summer uplift documented onsite.

- ▶ 26.5% total reduction in refrigeration energy usage
- ▶ 219+ kWh/day saved across freezer rooms alone
- ▶ 58 kWh/day saved across cold rooms
- ▶ 101,126 kWh saved on measured pilot basis
- ▶ 112,432 kWh full-year (seasonally adjusted)
- ▶ Sub-12-month payback achieved

VERIFIED METRICS

Technology	enPact R&C
Refrigeration reduction	26.5%
Annualized savings	112,432 kWh
CO ₂ abated	66,335 kg/yr
Trial duration	34 days (Feb–Mar)
Payback	~11 months

Measurement & verification. All equipment monitored via smart kWh meters feeding the Univers platform. Winter measurement was seasonally adjusted upward using the site's documented 25% summer refrigeration uplift to produce the full-year 112,432 kWh figure. No refrigerant, setpoint, or workflow changes were made during the trial.

CONCLUSION

enPact Refrigeration & Cooling delivered a validated 26.5% reduction across a large refrigerated warehouse, ~11-month payback, and material CO₂ abatement — with zero cold-chain disruption.

ESTATE ROLLOUT PROJECTION

Applied across 340 similar sites: 34.4M kWh/yr savings, 7,119 tons CO₂/yr abated, ~12-month portfolio payback.

Interested in achieving similar savings across your estate?

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